STPM 2016 PHYSICS 960/1

Questions and Answers

All calculations use $g = 9.81\ \text{m s}^{-2}$ and the values of constants printed in the question paper.

Section A [15 marks]

1. A car moves with a constant speed of 40 m s^{-1} towards east, and then moves with a constant speed of 30 m s^{-1} towards north. What is the change in velocity of the car?

A 10 m s^{-1} at 053° B 10 m s^{-1} at 037° C 50 m s^{-1} at 323° D 50 m s^{-1} at 307°

Answer: D

$$\Delta \boldsymbol{v} = \boldsymbol{v} - \boldsymbol{u} = (30\ \text{m s}^{-1}\ \text{north}) + (40\ \text{m s}^{-1}\ \text{west})$$

$$|\Delta v| = \sqrt{(30^2 + 40^2)} = 50\ \text{m s}^{-1}$$

$$\text{Angle west of north: } \tan^{-1}\frac{40}{30} = 53.1° \Longrightarrow \text{bearing} = 360° - 53.1° = 307°$$

Explanation: Change in velocity is final velocity minus initial velocity, so add $-\boldsymbol{u}$ (40 m s^{-1} towards the west). Bearings are measured clockwise from north.

2. A bus is moving at a constant velocity of 20 m s^{-1}. Just as the bus passes a stationary car, the car starts to accelerate at 2.0 m s^{-2}. What is the distance travelled by the car when it overtakes the bus?

A 100 m B 200 m C 300 m D 400 m

Answer: D

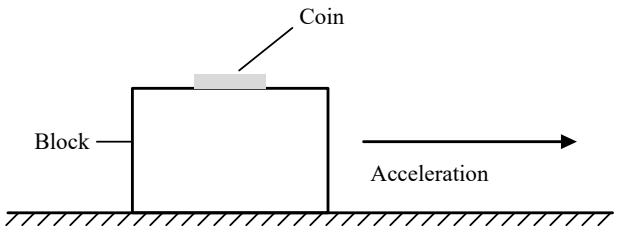
$$\text{Bus: } s = 20t\text{; car: } s = \frac{1}{2}(2.0)t^2 = t^2$$

$$t^2 = 20t \Longrightarrow t = 20\ \text{s}$$

$$s = 20 \times 20 = 400\ \text{m}$$

Explanation: When the car overtakes the bus they have travelled the same distance in the same time. Set the two displacement expressions equal.

3. A coin is placed on top of a block as shown in the diagram below.



If the coefficient of static friction between the coin and the block is 0.30, what is the maximum acceleration of the block so that the coin does **not** slip?

A 0.30 m s^{-2} B 2.9 m s^{-2} C 9.8 m s^{-2} D 33 m s^{-2}

Answer: B

Friction provides the coin's acceleration: $\mu mg = ma$

$$a_{max} = \mu g = (0.30)(9.81) = 2.9 \text{ m s}^{-2}$$

Explanation: The only horizontal force on the coin is static friction. If the block accelerates faster than μg, friction cannot accelerate the coin enough and the coin slips.

4. A car is travelling on a banked road at an angle of 20° to the horizontal. If the horizontal radius of the road is 50.0 m and the frictional force is negligible, what is the speed of the car so that it does not skid?

A 13 m s^{-1} B 22 m s^{-1} C 31 m s^{-1} D 37 m s^{-1}

Answer: A

Vertical: $R \cos 20° = mg$; horizontal: $R \sin 20° = \frac{mv^2}{r}$

$$\tan 20° = \frac{v^2}{rg} \Longrightarrow v = \sqrt{((50.0)(9.81) \tan 20°)} = 13 \text{ m s}^{-1}$$

Explanation: With no friction, the horizontal component of the normal reaction provides the centripetal force. The speed depends only on the angle of banking and the radius.

5. A satellite of mass m goes around the Earth of mass M in an orbit of radius r. If the satellite moves to a new orbit of radius $\frac{r}{2}$, which is true about the total energy of the satellite?

A Decreases by $\frac{3GmM}{r^2}$ B Increases by $\frac{3GmM}{r^2}$

C Decreases by $\frac{GmM}{2r}$ D Increases by $\frac{GmM}{2r}$

Answer: C

$$E = -\frac{GMm}{2r}; \text{ at radius } \frac{r}{2}: E = -\frac{GMm}{r}$$

$$\Delta E = -\frac{GMm}{r} + \frac{GMm}{2r} = -\frac{GMm}{2r}$$

The total energy decreases by $\frac{GMm}{2r}$.

Explanation: The total energy of an orbiting satellite is negative and equal to half its potential energy. A smaller orbit has a more negative total energy, even though the kinetic energy increases.

6. The period of a satellite in a circular orbit depends on

A the mass of the satellite
B the direction of rotation of the satellite
C the distance of the satellite from the surface of the Earth
D the speed of the satellite launched from the surface of the Earth

Answer: C

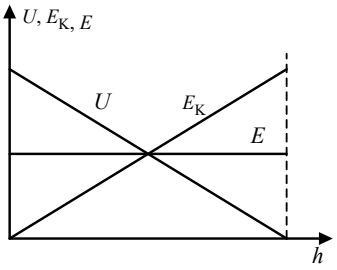
$$T^2 = \frac{4\pi^2}{GM}(R + h)^3$$

The period depends on the orbit radius, i.e. on the distance h of the satellite from the Earth's surface.

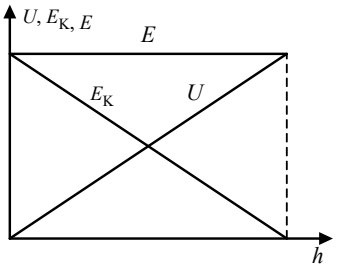
Explanation: The period does not depend on the mass of the satellite or its direction of rotation. The speed in orbit is fixed by the radius, not by the launch speed.

7. A stone is thrown vertically upward under gravitational attraction. Which graph represents the variation of potential energy U, the kinetic energy E_K and the total energy E of the stone with its height h?
[Assume that air resistance is negligible.]

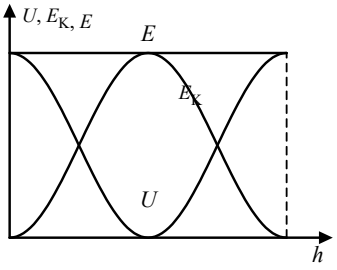
A



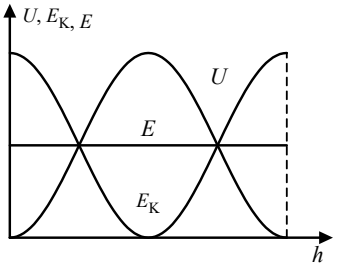
B



C



D



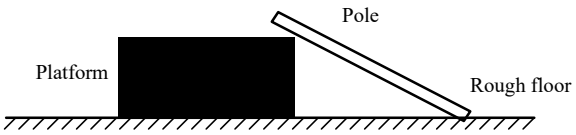
Answer: B

$U = mgh$ increases linearly with h; $E_K = E - mgh$ decreases linearly; E is constant

At the maximum height $U = E$ and $E_K = 0$, so E is the horizontal line at the top of the graph.

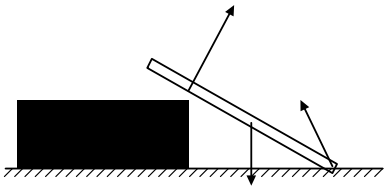
Explanation: The graph is against height, not time, so the lines are straight. Total energy equals the maximum value of either U or E_K, so it cannot cross them in the middle (graph A).

8. A uniform pole, leaning against an edge of a platform, is shown in the diagram below.

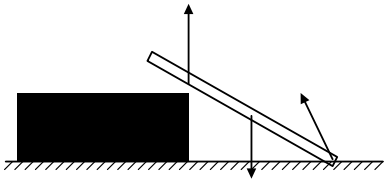


If the pole is in equilibrium, which diagram shows the direction of forces acting on the pole?

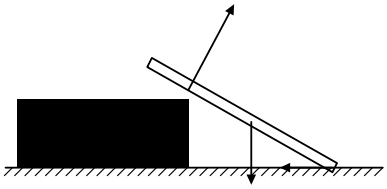
A



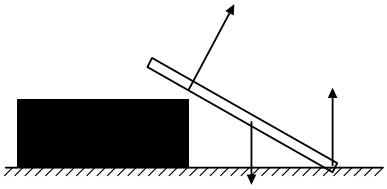
B



C



D



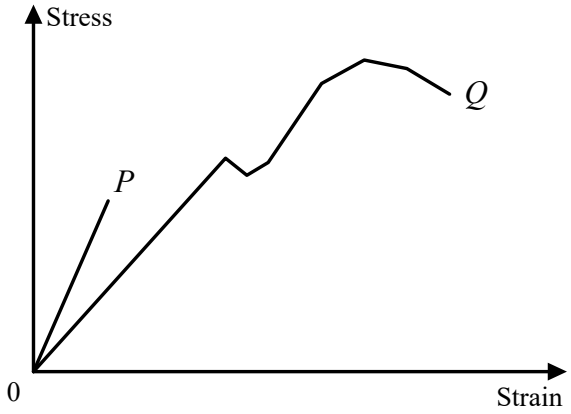
Answer: A

The forces are: the weight vertically downward at the centre, the reaction at the platform edge perpendicular to the pole, and the force at the rough floor (normal reaction + friction) directed upward and towards the platform.

For equilibrium the three forces must pass through one point and form a closed triangle, as in A.

Explanation: The edge of the platform is smooth contact, so its force is perpendicular to the pole. The floor force must have a horizontal component (friction) to balance the horizontal component of the edge reaction.

9. The variation of stress with strain for materials *P* and *Q* until their breaking points are shown in the graph below.



Which is true about materials *P* and *Q*?

A Material *Q* is stiffer than material *P*.
B Material *P* is ductile but material *Q* is brittle.
C Material *Q* is more difficult to break compared to material *P*.
D Material *P* needs more work to break compared to material *Q*.

Answer: C

Q reaches a higher stress before it breaks, so *Q* is more difficult to break than *P*.

Explanation: *P* has the steeper gradient (stiffer, so A is wrong) and breaks with no plastic region (brittle, so B is wrong). The area under the curve of *Q* is much larger, so *Q* needs more work to break (D is wrong).

10. Which statement is **not** about an assumption in kinetic theory for an ideal gas?

A The loss of kinetic energy during collisions is negligible.
B The change of momentum during collision with the wall is negligible.
C The time of collision is negligible compared to the time between collisions.
D The volume of molecules is negligible compared to the volume of the container.

Answer: B

The change of momentum of a molecule on hitting the wall is $2mc_x$; it is not negligible. It is this change of momentum that produces the pressure, so B is not an assumption.

Explanation: The other statements are standard assumptions: elastic collisions, negligible collision time and negligible molecular volume.

11. Which statement is true about degrees of freedom of a gas molecule?

A The total degrees of freedom of translational motion of a monatomic gas molecule at room temperature is 1.
B The total degrees of freedom of rotational motion of a monatomic gas molecule at room temperature is 3.
C The total degrees of freedom of translational motion of a diatomic gas molecule at room temperature is 5.
D The total degrees of freedom of rotational motion of a diatomic gas molecule at room temperature is 2.

Answer: D

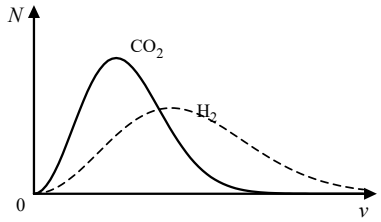
A diatomic molecule at room temperature has 3 translational and 2 rotational degrees of freedom.

Explanation: A monatomic molecule has 3 translational and no (significant) rotational degrees of freedom, so A, B and C are wrong.

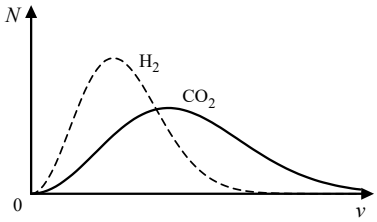
12. Which graph represents the distribution of the molecular speed of the same numbers of molecules of hydrogen (H_2) and carbon dioxide (CO_2) gases at the same temperature?

[Ndv represents the number of molecules with speed in the range v to v + dv.]

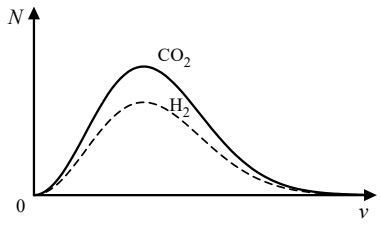
A



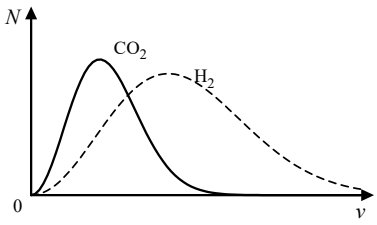
B



C



D



Answer: A

$c_{\text{r.m.s.}} = \sqrt{(\frac{3RT}{M})}$: CO_2 ($M = 44$ g mol^{-1}) is much heavier than H_2 (2 g mol^{-1})

So the CO_2 distribution peaks at a lower speed and is narrower and taller, while H_2 is spread to higher speeds. Both curves enclose the same area (same number of molecules).

Explanation: At the same temperature both gases have the same average kinetic energy, so the lighter molecules move faster. Equal numbers of molecules means equal areas under the two curves.

13. The molar heat capacity at constant pressure of an ideal gas is $C_{\text{p,m}}$. If R is the molar gas constant, which expression represents the number of degrees of freedom of the gas?

A $\frac{2C_{\text{p,m}}}{R}$ B $\frac{C_{\text{p,m}}}{R} - 1$ C $2(\frac{C_{\text{p,m}}}{R} - 1)$ D $2(1 - \frac{C_{\text{p,m}}}{R})$

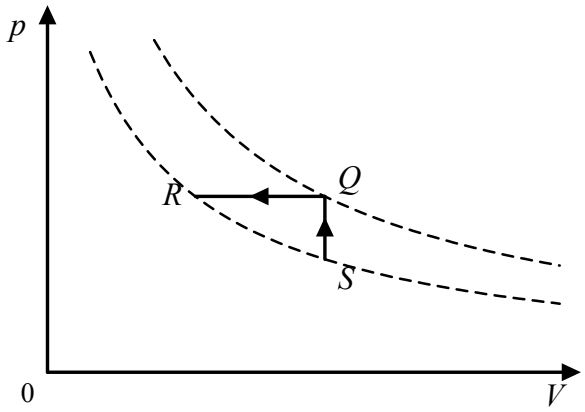
Answer: C

$$C_{p,m} = C_{V,m} + R = \frac{f}{2}R + R$$

$$f = 2(\frac{C_{p,m}}{R} - 1)$$

Explanation: Each degree of freedom contributes $\frac{1}{2}R$ to $C_{V,m}$, and $C_{p,m} - C_{V,m} = R$. Rearrange for f.

14. The isothermal curves of a fixed mass of an ideal gas which undergoes changes *SQ* and *QR* are shown in the *p*-*V* graph below.



Which statement describes the changes correctly?

A Heat is supplied to the gas in process *QR*.
B Work is done when the gas is heated from *S* to *Q*.
C The internal energy of the gas in the whole process increases.
D The heat supplied in process *SQ* increases the internal energy of the gas.

Answer: D

SQ is at constant volume (no work done) and the gas moves to a higher isotherm, so its temperature rises: the heat supplied increases the internal energy.

QR is a compression at constant pressure back to the lower isotherm: heat is released.

Explanation: *S* and *R* are on the same isotherm, so the internal energy over the whole process is unchanged (C wrong). No work is done in *SQ* (B wrong).

15. Which is **not** a way to reduce global warming?

A Use more fossil fuel
B Use more public transport
C Use more renewable energy
D Use more energy saving electrical appliances

Answer: A

Burning more fossil fuel releases more carbon dioxide, a greenhouse gas, so it increases global warming.

Explanation: Public transport, renewable energy and energy-saving appliances all reduce the burning of fossil fuels.

Section B [15 marks]

16. (a) State the conditions for equilibrium of a rigid body. [2]

Answer

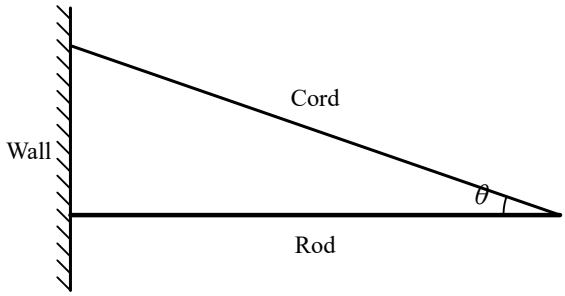
The resultant force acting on the body is zero — 1

The resultant torque about any axis (point) is zero — 1

2

Explanation: The first condition prevents linear acceleration; the second prevents angular acceleration.

(b) A uniform rod of length L and mass m is held horizontally against a vertical wall by a light cord making an angle θ with the rod as shown in the diagram below.



The coefficient of static friction between the end of the rod and the wall is 0.42. Determine the maximum value for θ if the rod is to remain in equilibrium. [5]

Answer

Moments about the end at the wall: $(T \sin\theta)L = mg\frac{L}{2} \Longrightarrow T\sin\theta = \frac{mg}{2}$ — 1

Vertical: $f + T\sin\theta = mg \Longrightarrow$ friction $f = \frac{mg}{2}$ — 1

Horizontal: $N = T\cos\theta = \frac{mg}{2\tan\theta}$ — 1

For no slipping: $f \le \mu N \Longrightarrow \frac{mg}{2} \le \mu\frac{mg}{2\tan\theta} \Longrightarrow \tan\theta \le \mu$ — 1

$\theta_{max} = \tan^{-1} 0.42 = 22.8°$ — 1

5

Total for question **7**

Explanation: As θ increases, the cord pulls less horizontally, so the normal reaction (and the maximum friction) at the wall decreases, while the friction needed stays $\frac{mg}{2}$. The rod slips when the friction needed exceeds μN.

17. (a) Deduce from the first law of thermodynamics the change in internal energy of a gas after undergoing an adiabatic process. [3]

Answer

Adiabatic: no heat enters or leaves the system, $Q = 0$	1
$Q = \Delta U + W \Longrightarrow \Delta U = -W$	1
In an adiabatic expansion (W positive) the internal energy decreases; in an adiabatic compression it increases	1
	3

Explanation: With no heat exchanged, the work done by the gas comes entirely from its internal energy, so the gas cools as it expands.

(b) A diatomic gas with $\gamma = 1.40$ initially at a temperature of 250 K, a volume of 2.35×10^{-2} m^3 and a pressure of 150 kPa undergoes an adiabatic expansion to a volume of 4.00×10^{-2} m^3. Determine the

(i) final temperature of the gas, [2]

Answer

$T_1V_1^{\gamma-1} = T_2V_2^{\gamma-1} \Longrightarrow T_2 = 250\left(\frac{2.35 \times 10^{-2}}{4.00 \times 10^{-2}}\right)^{0.40}$	1
$T_2 = 202$ K	1
	2

Explanation: Use the adiabatic relation in terms of temperature and volume.

(ii) work done by the gas. [3]

Answer

$nR = \frac{p_1V_1}{T_1} = \frac{(150 \times 10^3)(2.35 \times 10^{-2})}{250} = 14.1$ J K^{-1}	1
$W = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{14.1(250 - 202.1)}{0.40}$	1
$W = 1.69 \times 10^3$ J	1
	3
Total for question	**8**

Explanation: The work done by the gas equals the decrease in internal energy, $nC_{V,m}(T_1 - T_2)$, with $C_{V,m} = R/(\gamma - 1)$.

Section C [30 marks]

18. (a)

(i) Define *acceleration*. [1]

Answer

Acceleration is the rate of change of velocity — 1

1

Explanation: $a = \mathrm{d}v/\mathrm{d}t$; it is a vector quantity.

(ii) Derive an equation of motion with the initial velocity u, the final velocity v, and the displacement s at a constant acceleration a. [2]

Answer

$s = \frac{1}{2}(u + v)t$ and $t = \frac{v - u}{a} \Longrightarrow s = \frac{1}{2}(u + v)\frac{v - u}{a}$ — 1

$2as = v^2 - u^2 \Longrightarrow v^2 = u^2 + 2as$ — 1

2

Explanation: For constant acceleration the average velocity is $\frac{1}{2}(u + v)$. Eliminating t gives the equation without time.

(b) A driver is driving a car with a velocity of 72 km h^{-1}. Suddenly a child runs across the road at a distance of 30.0 m in front of his car and he brakes immediately. The driver's response time is 0.50 s.

(i) Determine the minimum deceleration of the car so that the child will not get knocked down. [3]

Answer

$u = 72$ km $h^{-1} = 20$ m s^{-1}; distance during the response time $= (20)(0.50) = 10$ m — 1

Braking distance available $= 30.0 - 10 = 20$ m — 1

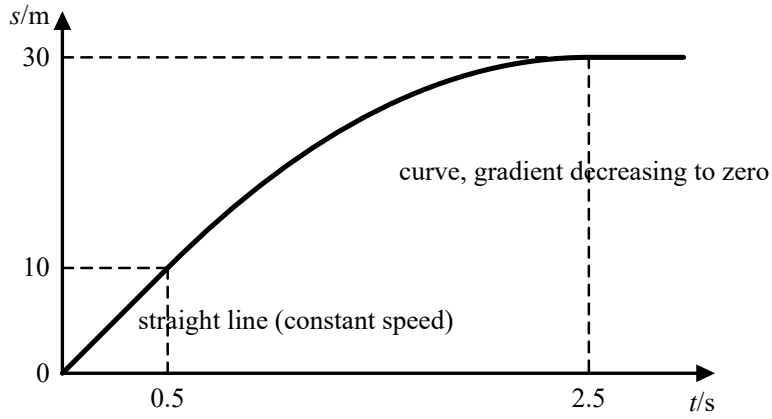
$0 = 20^2 + 2a(20) \Longrightarrow a = -10$ m s^{-2}; minimum deceleration $= 10$ m s^{-2} — 1

3

Explanation: During the response time the car still moves at constant speed. Only the remaining distance is available for braking.

(ii) Sketch a graph of displacement against time for the motion of the car. [2]

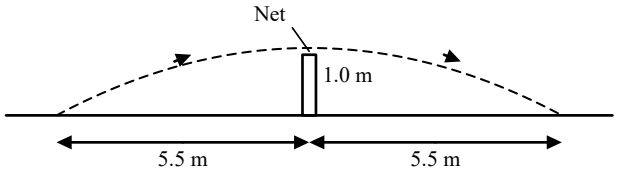
Answer



Straight line (constant gradient) for the first 0.5 s, reaching 10 m	1
Curve with decreasing gradient, becoming horizontal at 30 m (at 2.5 s)	1
	2

Explanation: The gradient of a displacement–time graph is the velocity: constant during the response time, then decreasing to zero as the car stops just at the child.

(c) A tennis ball being hit over the net is shown in the diagram below.



(i) Determine the vertical component of the initial velocity of the tennis ball. [2]

Answer

The ball is at its highest point (1.0 m) above the net, where $v_y = 0$: $0 = u_y^2 - 2(9.81)(1.0)$	1
$u_y = 4.43$ m s^{-1}	1
	2

Explanation: The path is symmetrical (5.5 m on each side of the net), so the maximum height is reached directly above the net.

(ii) Calculate the time taken for the tennis ball to land. [2]

Answer

$0 = 4.43t - \frac{1}{2}(9.81)t^2$ 1

$t = 0.903$ s 1

2

Explanation: The ball lands at the same level as it was hit, so the vertical displacement is zero. The time is twice the time to reach the top.

(iii) Determine the angle of projection of the tennis ball above the horizontal. [3]

Answer

$u_x = \frac{5.5 \times 2}{0.903} = 12.2 \text{ m s}^{-1}$ 1

$\tan\theta = \frac{u_y}{u_x} = \frac{4.43}{12.18} = 0.364$ 1

$\theta = 20.0°$ 1

3

Total for question **15**

Explanation: The horizontal velocity is constant, so it is the total range (11.0 m) divided by the time of flight.

19. (a)

(i) Explain the condition when a body which is in motion can have a change in velocity and at the same time have constant acceleration. [3]

Answer

A body in uniform circular motion moves with constant speed but its direction keeps changing, so its velocity changes 1

It has an acceleration because a (centripetal) force acts on it towards the centre 1

The magnitude of the acceleration, v^2/r, is constant (always directed towards the centre) 1

3

Explanation: Constant acceleration here means constant magnitude; its direction keeps changing so that it always points to the centre.

(ii) Derive the relationship between the velocity and the acceleration in (a)(i). [4]

Answer

In a short time dt the velocity turns through $d\theta$; from the vector triangle $|d\boldsymbol{v}| = v\,d\theta$ — 1

$a = \dfrac{dv}{dt} = v\dfrac{d\theta}{dt} = v\omega$ — 1

$s = r\theta \Longrightarrow v = r\dfrac{d\theta}{dt} = r\omega$, so $\omega = \dfrac{v}{r}$ — 1

$a = \dfrac{v^2}{r}$ — 1

4

Explanation: The velocity vectors at two nearby points have the same magnitude v; the small change between them points towards the centre.

(b) A communication satellite orbits the Earth at a fixed distance h from the surface of the Earth with a speed of 6500 m s^{-1}.

(i) Calculate h. [4]

Answer

$\dfrac{GMm}{(R+h)^2} = \dfrac{mv^2}{R+h}$ — 1

$R + h = \dfrac{GM}{v^2} = \dfrac{(6.67 \times 10^{-11})(5.97 \times 10^{24})}{6500^2}$ — 1

$R + h = 9.42 \times 10^6$ m — 1

$h = 9.42 \times 10^6 - 6.38 \times 10^6 = 3.04 \times 10^6$ m — 1

4

Explanation: The orbit radius is measured from the centre of the Earth, so subtract the Earth's radius to get the height above the surface.

(ii) Determine the period T of the satellite motion. [2]

Answer

$T = \dfrac{2\pi(R+h)}{v} = \dfrac{2\pi(9.42 \times 10^6)}{6500}$ — 1

$T = 9.11 \times 10^3$ s — 1

2

Explanation: One orbit is a distance $2\pi(R + h)$ at constant speed.

(c) Why can the satellite in (b) stay in orbit during its life span? [2]

Answer

The gravitational force of the Earth on the satellite provides exactly the centripetal force needed for its circular orbit	1
Its speed matches the orbit radius and, at that height, air resistance is negligible, so it does not lose speed	1
	2
Total for question	**15**

Explanation: The satellite is continuously falling towards the Earth, but its tangential speed carries it around the Earth so it never gets closer.

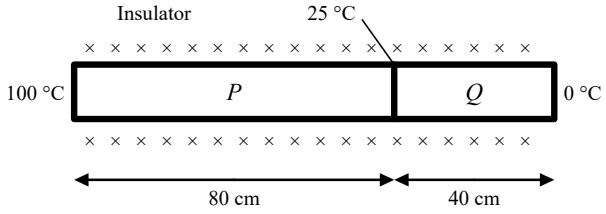
20. (a) Describe **two** mechanisms for thermal conduction in a metal. [4]

Answer

Lattice vibration: atoms at the hot end vibrate with greater amplitude	1
and pass the energy to neighbouring atoms through the bonds (collisions), towards the colder end	1
Free electron diffusion: free electrons at the hot end gain kinetic energy and diffuse rapidly to the colder end	1
where they transfer energy to the atoms by colliding with them	1
	4

Explanation: In metals the free-electron mechanism is much faster, which is why metals are good conductors.

(b) Metal rods *P* and *Q* of the same cross-sectional area, of length 80 cm and 40 cm respectively, are joined as shown in the diagram below.



The thermal conductivities of metal rods P and Q are k_P and k_Q respectively. Both rods are well insulated.

(i) Write an expression for the rate of heat flow in the rod. [2]

Answer

$\frac{dQ}{dt} = -kA\frac{d\theta}{dx}$ 1

k = thermal conductivity, A = cross-sectional area, $d\theta/dx$ = temperature gradient 1

2

Explanation: The negative sign shows that heat flows in the direction of decreasing temperature.

(ii) State **two** conditions for the expression in (b)(i) to be valid. [2]

Answer

• The steady state has been reached (temperatures no longer change with time) 1

• The rods are perfectly insulated so no heat is lost from the sides (heat flow is parallel to the rod) 1

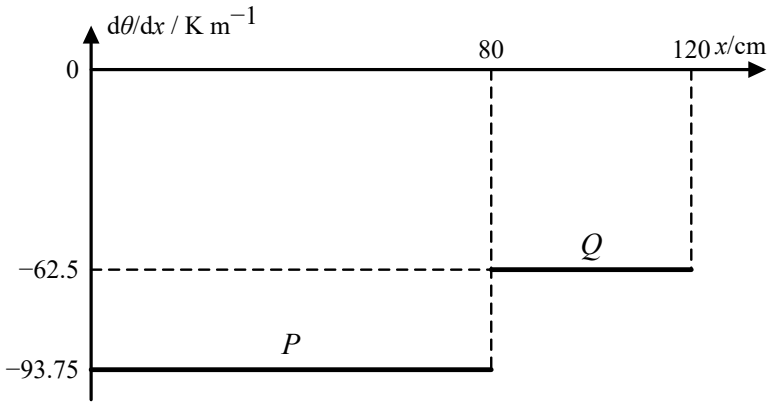
2

Explanation: Under these conditions the rate of heat flow is the same at every cross-section of the joined rods.

(iii) Sketch a graph of temperature gradient against distance for the joined rods. Which rod is a better thermal conductor? Explain your answer. [5]

Answer

Gradient in $P = \frac{25 - 100}{0.80} = -93.75$ K m^{-1}; in $Q = \frac{0 - 25}{0.40} = -62.5$ K m^{-1} 2



Correct graph: two horizontal (constant) lines, with values 1

Q is the better thermal conductor 1

Same rate of heat flow and area, so $k \propto 1/|d\theta/dx|$; Q has the smaller temperature gradient 1

5

Explanation: In each rod the temperature falls linearly, so the temperature gradient is constant within each rod and changes at the junction.

(iv) Determine the value of $\frac{k_P}{k_Q}$. [2]

Answer

$k_P(93.75) = k_Q(62.5) \Longrightarrow \frac{k_P}{k_Q} = \frac{62.5}{93.75}$ 1

$\frac{k_P}{k_Q} = 0.67$ 1

2

Total for question **15**

Explanation: In the steady state the same rate of heat flows through both rods of equal cross-section.